

Feather Key

and permit relative axial movement of the pair. It may be rectangular, square, dovetail (or) rounded in cross-section.

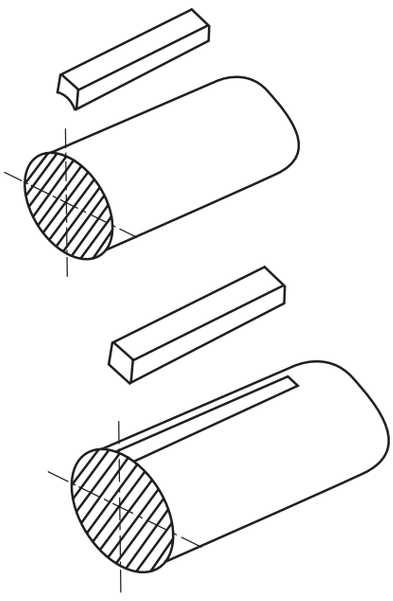
Saddle Key

The top of this key is fixed into the keyway of the hub. The bottom of the key is mounted on the shaft. There is no keyway in the shaft. Saddle keys are used for low power transmission.

There are two types of saddle key

1. Hollow Saddle Key
2. Flat Saddle Key

Hollow Saddle Key: A hollow saddle key has a concave shaped bottom to suit the curved surface of the shaft on which it is used. The keyway is only cut in the hub of the wheel. The relative rotation is prevented by the friction between the key and the shaft.



Hollow and Flat Saddle Key

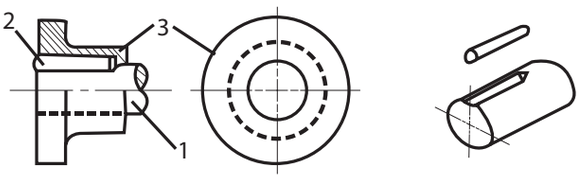
Flat Saddle Key: It is similar to the hollow saddle key except that the bottom surface of it is flat. It fits on the flat surface provided on the shaft. It gives more gripping than a hollow saddle key.

Round Key

Keys of circular cross-section are called rounded keys, usually tapered along the length. A round key fits in the hole drilled partly in the shaft and partly in the hub. It is generally used for light duty transmission of power.

There are two types of round keys.

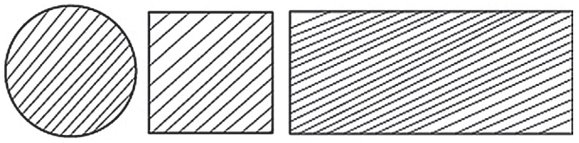
1. Tapered Round Key
2. Plain Round Key



1. Shaft 2. Round Key 3. Hub

Tapered and Plain Round Key

Faces of Keys: There are three types of key faces. The cross-sectional view of the key is as follows



Round Square Rectangle

BME_Chp07.indd 94 11/9/2020 12:24:38 PM